Thesis roadmap: physical understanding through interaction

STEP 1 · PART 0 | **STEP 2 · PART I** | STEP 3 · PART II | STEP 4 · PART III | STEP 5 · PART IV

Control under a given model

Grounded concept & skill formation

Causal rules of change

Active testing & model revision

Minimal, transferable representations

External environment sensors + real-time control

Longer-term thesis goals

NR

Material evolution + future command / control

Interaction history → **Physical state** → Future response

Force, motion and past commands

What the material retains

Force and motion after a new action

Planned testbed: a robot arm with force and motion sensing.

LT (parallel): relational programs, version-space inference and active evidence acquisition.

What should a robot retain from one interaction to predict the next?

[1]

Policy reuse and the current research question

Original goal: policy reuse

Can a previously learned pressing policy
still work when the conditions change?

Current research question

Can we infer the current physical state from interaction history
to assess whether a learned policy can be reused?

Research interface: infer the material state,
then predict the response of a candidate policy.

[1]

Physical state across repeated contact

Motivating interaction



Releasing contact does not necessarily reset the material.

Similar current force and position may conceal different internal recovery states.

High-level idea

Ground physical state in interaction history, then use its evolution to predict the consequences of a new action.

1 Grounding

Does history resolve ambiguity in current readings?

2 Transfer

Can the same state predict under new commands and control settings?

3 Action value

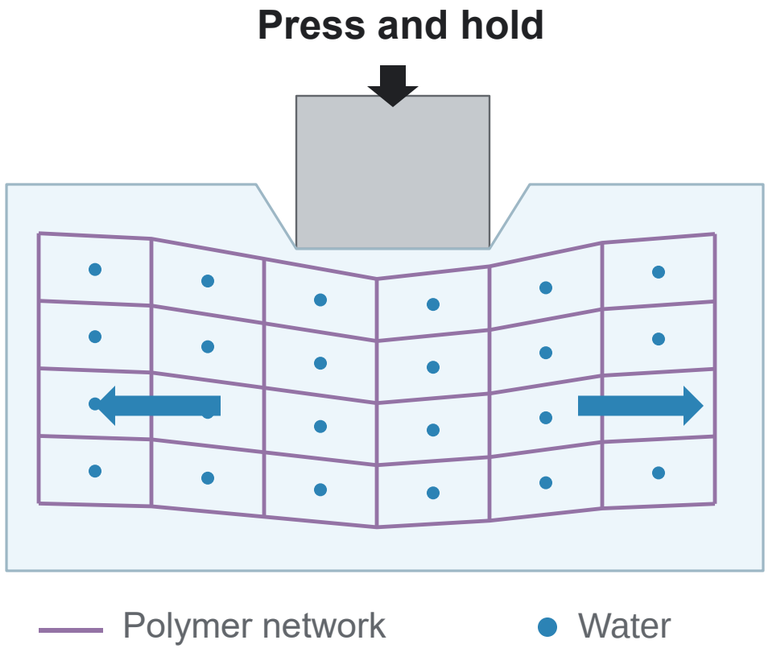
Does this information improve actions when online feedback is available?

First test: different histories, matched current readings, identical future command.

[4–6] [E1–E3]

Pressing a water-rich gel

A soft polymer network holds water. One press changes both.



Conceptual view inside the gel

Internal process	What changes during the hold
Faster in this study **Network relaxation**	Polymer chains rearrange. Stretched parts of the network relax.
Slower in this study **Water redistribution**	Water moves through small pores. Pressure redistributes within the gel.

PEG hydrogel indentation experiments
Bush et al., Soft Matter, 2015 [8]

The gel keeps changing internally while the indenter stays still.

Relative speeds depend on the material and contact size.

[8]

RME, CPF and the estimation target

All three use physical structure to infer quantities that are not directly observed.

	Evidence	Estimated quantity	Purpose
RME	Proprioceptive dynamics mismatch	Payload mass and center of mass	Adapt compliant execution
CPF	Momentum residual and contact constraints	External contact location and force	Localize physical contact
NR	Loading history and measured response	Evolving internal material state	Predict a changed future interaction

The extension is the evolving material state and its consequences across interactions.

[2, 3]

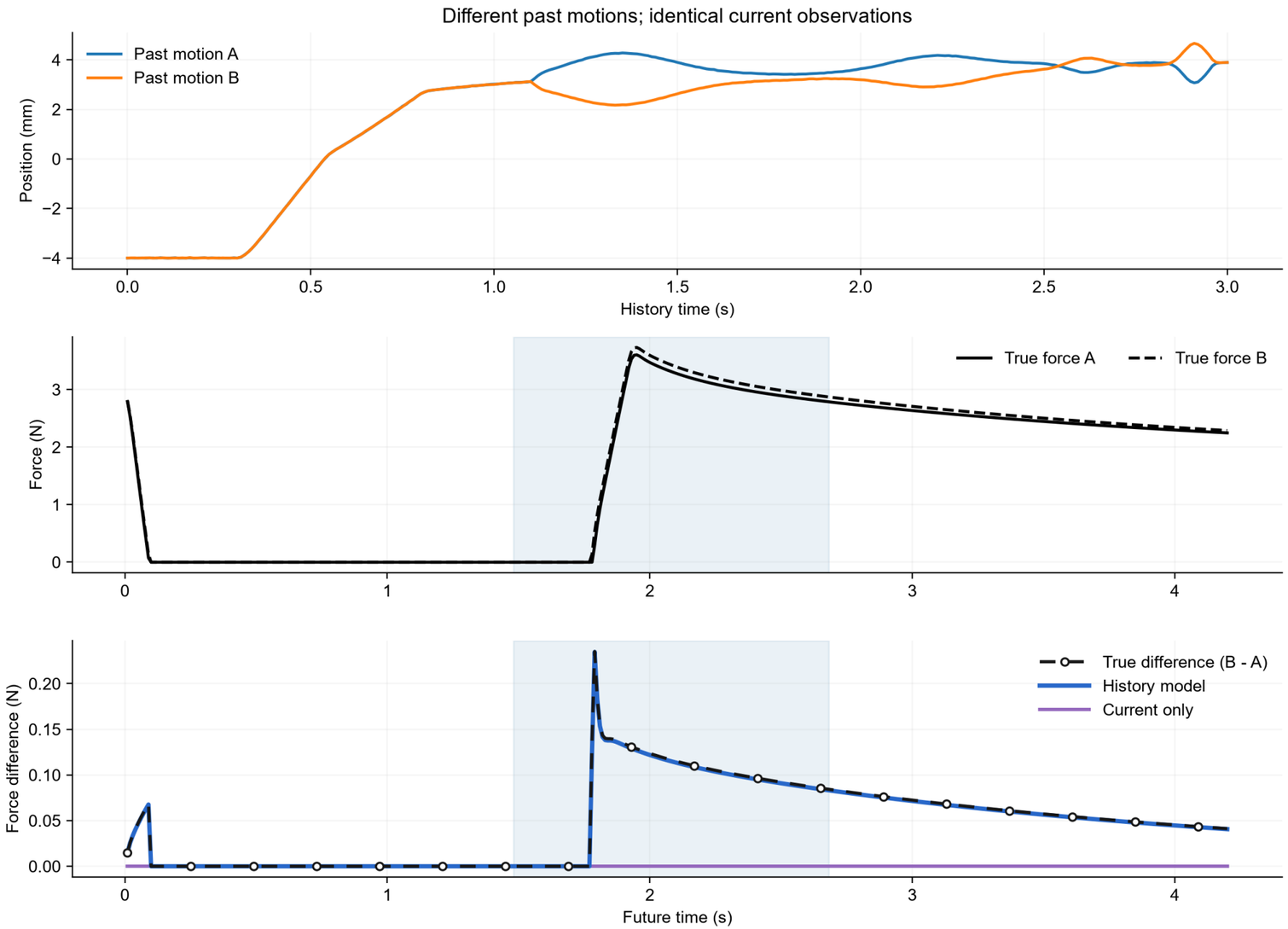
History-based state inference: related approaches

Prior work	Core idea	Relation to NR
POMDP / belief state Kaelbling et al., 1998	Update state probabilities from past actions and observations.	Foundation for retaining state and uncertainty.
PlaNet Hafner et al., 2019	Infer latent state from images and actions, then predict and plan.	Separate history-to-state inference from action-conditioned prediction.
RMA Kumar et al., 2021	Infer an environment encoding from recent state and action history.	Condition a reusable policy on an updated environment encoding.
Material internal variables Rosenkranz et al., 2024	Use an auxiliary LSTM in training a constrained material model.	Infer material memory from observed deformation and stress.

Design choice: infer a compact material state, then evolve it under the next action.

[4–7]

Material memory hidden by the current force



Fast and slow modes can give the same force now, then evolve differently.

History estimates their allocation to predict the next contact.

How the history model is trained

A GRU estimates the current material state. Physics predicts what happens next.

Step	What the model does
1 Read the history	Past 3 s, sampled every 0.01 s: position, velocity, contact force, command and time step.
2 Estimate the state	The GRU estimates the allocation between fast and slow components, giving the current material state (z1, z2).
3 Predict the response	Fixed dynamics use that state, current motion and planned commands to predict future force and position.
4 Train the GRU	Compare predictions with subsequent simulation records. Backpropagate force and position errors to update the GRU.

Material parameters are fitted on training data, then held fixed.

1D motion with a 2D material state.

History-based state estimation and prediction

1 Estimate the current state

H_t: 3 s of (x, v, F, u, Δt), sampled every 0.01 s

$$h_t = \mathrm{GRU}_\theta(H_t), \quad \alpha = \sigma(w^\top h_t + b)$$

$$S_t = \max(k_\Sigma x_t - F_t, \varepsilon)$$

$$\hat{z}_1 = \alpha S_t / k_1, \quad \hat{z}_2 = (1 - \alpha) S_t / k_2$$

α allocates the weighted internal state
between fast and slow components.

2 Predict force and motion

$$k_\Sigma = k_\infty + k_1 + k_2$$

$$F = \max(k_\Sigma x - k_1 z_1 - k_2 z_2, 0)$$

$$q(z) = (k_1 z_1 + k_2 z_2)/k_\Sigma$$

$$\dot{z}_i = [\max(x, q(z)) - z_i]/\tau_i$$

$$\dot{x} = v, \quad m\dot{v} = K(u - x) - Dv - F$$

u: planned position command
K, D: controller stiffness and damping
q(z): free-surface compression
m: actuator mass τ_i: relaxation time

Integrate from $(x_t, v_t, \hat{z}_1, \hat{z}_2)$
with the planned future commands.

3 Train with future responses

$$L = \langle ((\hat{F} - F)/s_F)^2 + ((\hat{x} - x)/s_x)^2 \rangle$$

Future force and position supervise the estimate.
Backpropagation updates the GRU and allocation head.

s_F and s_x normalize the prediction errors.

Material parameters stay fixed during encoder training. 1D motion, 2D material state.

[E1]

Baseline characteristics and the RME adaptation

Material models (rows 1–3) and state estimation (row 4) test different parts of NR.

Baseline / source	Key characteristics	Role in our experiment
Generalized Maxwell / Prony Park & Schapery, 1999 [B1]	Fixed relaxation modes; interpretable parameters. No neural network.	Classical material-response reference. Select mode count on validation data.
LSTM stress predictor Rosenkranz et al., 2023 [B2]	Learns sequence memory; strain + time step → stress. No prescribed constitutive law.	Compare direct sequence prediction with explicit physical state and dynamics.
Physics-augmented model Rosenkranz et al., 2024 [6]	Learns free energy and a dissipation potential. Thermodynamic structure; LSTM assists training.	Compare learned material evolution. Adapt the paper to a 1D material model.
RME-inspired estimator Jaszczuk & Figueroa, 2025 [2]	Neural prior + variational inference. Quantifies uncertainty in physical unknowns.	Adapt mass/CoM estimation to material state z. Use the same fixed material law.

Model comparison: same deformation paths. State comparison: same dynamics and observations.
Report force error; for the RME adaptation, also report update time and uncertainty calibration.

Paper adaptations are planned. Current NR retains 1D motion and a 2D material state.

[B1] IJSS 1999 [B2] IJNME 2023 [6] Computational Mechanics 2024 [2] CoRL 2025 | Full references in notes

History distinguishes material states hidden by the current readings

Different loading paths can end at the same measured position, speed and force. Their later responses still differ, and history helps predict that difference.

The inferred state can be reused under different future controls

Encode the past once, then use the same material state to predict responses under new controller settings. History inference and future control stay separate.

Shared feedback largely removes the added action benefit

With common online feedback, the two initializations lead to nearly the same actions and outcomes. Classical estimators also recover useful state in this material model.

Current setting: simulated 1D motion with a 2D material state.

[E1–E3]

A2. References

Idea sources

[1] Thesis narrative and NR scope, 2026

[2] Jaszczuk & Figueroa, CoRL 2025
Rapid Mismatch Estimation

[3] Manuelli & Tedrake, IROS 2016
Contact Particle Filter

[4] Kaelbling et al., 1998
Belief-state estimation

[5] Hafner et al., ICML 2019
PlaNet

[6] Rosenkranz et al., 2024
Physics-augmented viscoelasticity

[7] Kumar et al., RSS 2021
Rapid Motor Adaptation

[8] Bush et al., Soft Matter 2015
PEG hydrogel relaxation

Project evidence and plan

[E1] Matched histories
and state inference

[E2] Prediction across
future controllers

[E3] Frozen encoders
under shared feedback

[P1] First real-material
session plan